

CLUB club

2026-10-09

Some basic initial comments

Your sequencing reads (or other sequences, e.g., contigs from de novo assembly) have been aligned to a reference sequence by a tool like bowtie2. The result is in a SAM/BAM file. Now you want to put together a summary "consensus" sequence that represents what's in your data vis a vis the reference sequence.

The consensus sequence is a statistical summary

Especially with viral data, the idea that the genomes present in a sample all carry a "consensus" genome sequence is extremely simplistic.

Or even that any of them do.

This is pretty obvious, but is almost never mentioned, so it feels like a widespread misconception, even though the idea of viral quasispecies is well established.

■ A consensus sequence is like a mean

There may be no virus particle containing the consensus sequence. Just as with a set of numbers, the mean is not necessarily in the set. E.g. 3, 4, 7, 10 have a mean of 6. A consensus sequence is also a form of summary statistic.

This makes for a nice puzzle, BTW. As the size of a set of numbers increases, what happens to the probability that their mean is a member of the set? (Given some assumptions / model.)

■ You have to deal with variation in any case

Even outside the viral world, there will still frequently be variation in your sequence data. This can arise from misincorporation errors during PCR, mis-reading of optical data while sequencing, contamination, etc.

Reference bias

The choice of reference sequence to align to can have a major impact on your eventual consensus. It's important to remember that the reference may not be particularly close to what's in your data. If the genomes in your sample differ sufficiently from the reference in a region, no reads will map to that part of the reference. If you then call a consensus from the resulting SAM file, you will necessarily end up with a sub-optimal result.

Something to be aware of: It is possible with some consensus callers to tell it to use the reference if no reads map to a region. In general this is probably a bad idea.

If you have good coverage, there's no real reason to give the reference sequence to a consensus caller. A legitimate use of the reference can be to adjudicate between reads with low coverage and similar quality - you may want to call in favour of the reference base.

Note that Geneious puts a `?` into consensus sequences for regions of the reference with no matching reads. It is not clear whether your organism does not have that region at all or whether you were unlucky and molecules with sequences that do match the missing region were simply not sequenced.

So how do you call a consensus?

The simplest thing to do is to proceed "site" by site (i.e., column by column) through the reference genome positions. For each site you look at the nucleotide calls (the bases and their qualities) and "call" a decision. The sequence of calls gives you a consensus sequence.

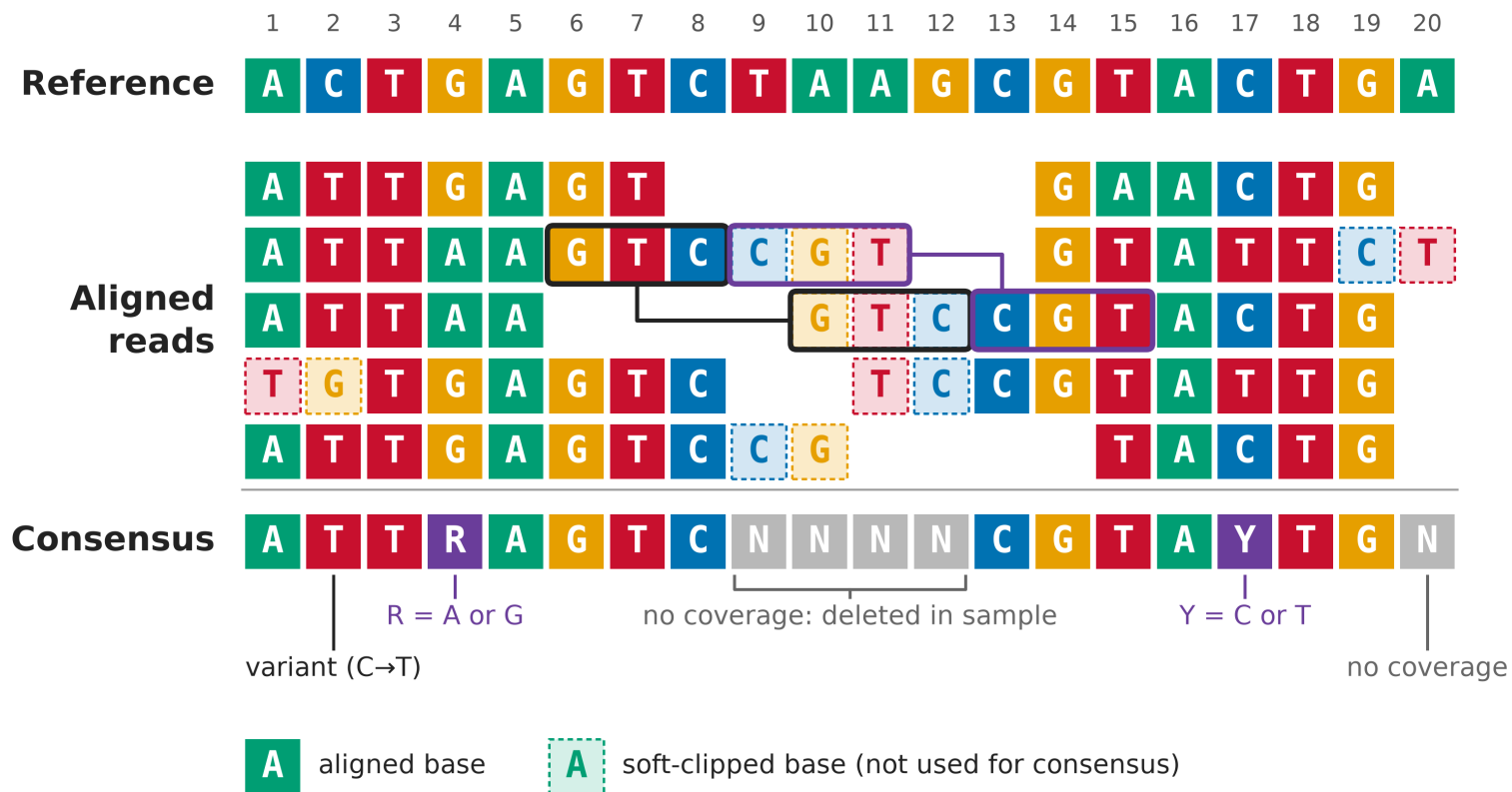
■ Obvious considerations

- How many reads will you require at minimum to make a call at a site?
- What homogeneity threshold(s) will you use?

■ Ambiguous nucleotide codes

M: AC
R: AG
W: AT
S: GC
K: GT
Y: CT
V: ACG
H: ACT
D: AGT
B: CGT
N: ACGT

An example



Upcoming classes

- The dark-matter tools.
- de novo assembly.
- Protein-level matching.
- Making multiple-sequence alignments.
- Tree making.